

Math-1730-04 Winter 2025 Information on the Exam

Document Date: March 31, 2025

Location: Toldo Health Education Centre 202

Time: Thursday, April 17, 3:30PM - 6:30PM

The Exam is a closed-book exam containing ten questions. It covers topics in Sections 5.3 - 5.5, 6.1 - 6.3, 6.5, 7.1 - 7.4, 7.8, 8.1, 8.2, 11.1 - 11.10.

Topics that you should know for the Exam include

- 1) Fundamental Theorem of Calculus - Part 1 and Part 2;
- 2) Substitution Rule, Integration by Parts, Trigonometric Substitution;
- 3) Trigonometric integrals, rational function integrals, improper integrals;
- 4) Area between curves, volume by disk/shell method, average value, arc length, surface area;
- 5) Convergence and divergence of sequences;
- 6) Convergence and divergence of series, Test for Divergence, conditional/absolute convergence, geometric series, telescoping series, p -series, alternating series and test;
- 7) Tests for positive series, for absolute convergence and for divergence: Comparison Test, Ratio Test, Root Test, Integral Test;
- 8) Power series, radius of convergence, interval of convergence;
- 9) Derivative and integral of a power series;
- 10) Power series representation of a function, Taylor series and Maclaurin series, calculation of the sum of a series by using Maclaurin series.

Formulas that you should know and can be used in your solutions include

- integral formulas from the Table in Section 5.4 (excluding $\sinh x$ and $\cosh x$);
- $\int \tan \theta d\theta = \ln |\sec \theta| + C$, $\int \sec \theta d\theta = \ln |\sec \theta + \tan \theta| + C$;
- the General Power Rule if you want (see the lecture outline of Section 5.5);
- the first six formulas on Maclaurin series from Table 1 in Section 11.10.

Known series that can be used for comparison in your solutions include

- geometric series and p -series.

Formulas provided during the Exam (see page 3) include

- some trigonometric identities and some formulas on applications of integration.

Hyperbolic and inverse hyperbolic functions cannot be used in your solutions.

Only non-programmable and non-graphic calculators are permitted. No other electronic device can be used during the Exam.

SAMPLES OF SHOWING WORK AND WRITING ANSWERS IN EXAM

Question 1. Find $\int \frac{1}{x(1 + (\ln x)^2)} dx$. Show your work and fill the black line with your answer.

Your answer: $\tan^{-1}(\ln x) + C$

Solution. Let $u = \ln x$. Then $du = \frac{1}{x} dx$. Therefore, we have

$$\int \frac{1}{x(1 + (\ln x)^2)} dx = \int \frac{1}{1 + (\ln x)^2} \frac{1}{x} dx = \int \frac{1}{1 + u^2} du = \tan^{-1} u + C = \tan^{-1}(\ln x) + C.$$

Question 2. Determine whether the series $\sum_{n=1}^{\infty} \frac{n^4 - n^{5/2} + 1}{2n^9 + 3}$ is convergent or divergent.

Show your reasoning and indicate the test that is used in your solution.

Your answer: The series is convergent.

Solution. Let $a_n = \frac{n^4 - n^{5/2} + 1}{2n^9 + 3}$ and $b_n = \frac{1}{n^5}$. Then

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{n^9 - n^{15/2} + n^5}{2n^9 + 3} = \lim_{n \rightarrow \infty} \frac{1 - n^{-3/2} + n^{-4}}{2 + 3n^{-9}} = \frac{1 - 0 + 0}{2 + 0} = \frac{1}{2} > 0.$$

Since $\sum_{n=1}^{\infty} b_n$ is the p -series with $p = 5$, which is convergent, by the Comparison Test, the given series is convergent.

Formula Sheet

Trigonometric Identities	Applications of Integration
$\cos^2 \theta + \sin^2 \theta = 1$	Area between curves: $\int_a^b (f(x) - g(x)) dx$
$1 + \tan^2 \theta = \sec^2 \theta$	Volumes by cross-sections: $\int_a^b A(x) dx$
$1 + \cot^2 \theta = \csc^2 \theta$	Volume by disk method: $\int_a^b \pi(\text{radius})^2 d(\text{variable})$
$\sin(x + y) = \sin x \cos y + \cos x \sin y$	Volume by washer method: $\int_a^b \pi [(\text{outer radius})^2 - (\text{inner radius})^2] d(\text{variable})$
$\sin(x - y) = \sin x \cos y - \cos x \sin y$	Volume by cylindrical shell method: $\int_a^b 2\pi(\text{variable})(\text{shell height}) d(\text{variable})$
$\cos(x + y) = \cos x \cos y - \sin x \sin y$	Arc length: $\int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$
$\cos(x - y) = \cos x \cos y + \sin x \sin y$	Area of surface of revolution (x -axis): $\int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$
$\sin 2\theta = 2 \sin \theta \cos \theta$	$\int_c^d 2\pi y \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$
$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$	Area of surface of revolution (y -axis): $\int_a^b 2\pi x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$
$\cos 2\theta = 2 \cos^2 \theta - 1$	$\int_c^d 2\pi x \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$
$\cos 2\theta = 1 - 2 \sin^2 \theta$	Average value of function: $\frac{1}{b-a} \int_a^b f(x) dx$
$\tan(x + y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$	
$\tan(x - y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}$	
$\sin x \cos y = \frac{1}{2} (\sin(x - y) + \sin(x + y))$	
$\sin x \sin y = \frac{1}{2} (\cos(x - y) - \cos(x + y))$	
$\cos x \cos y = \frac{1}{2} (\cos(x - y) + \cos(x + y))$	